



Department of Electrical and Electronic Engineering

Name of the Experiment: Verification of Ohm's Law

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26/11/25

Theory:

The current in a circuit is directly proportional to the applied voltage and inversely proportional to the resistance.

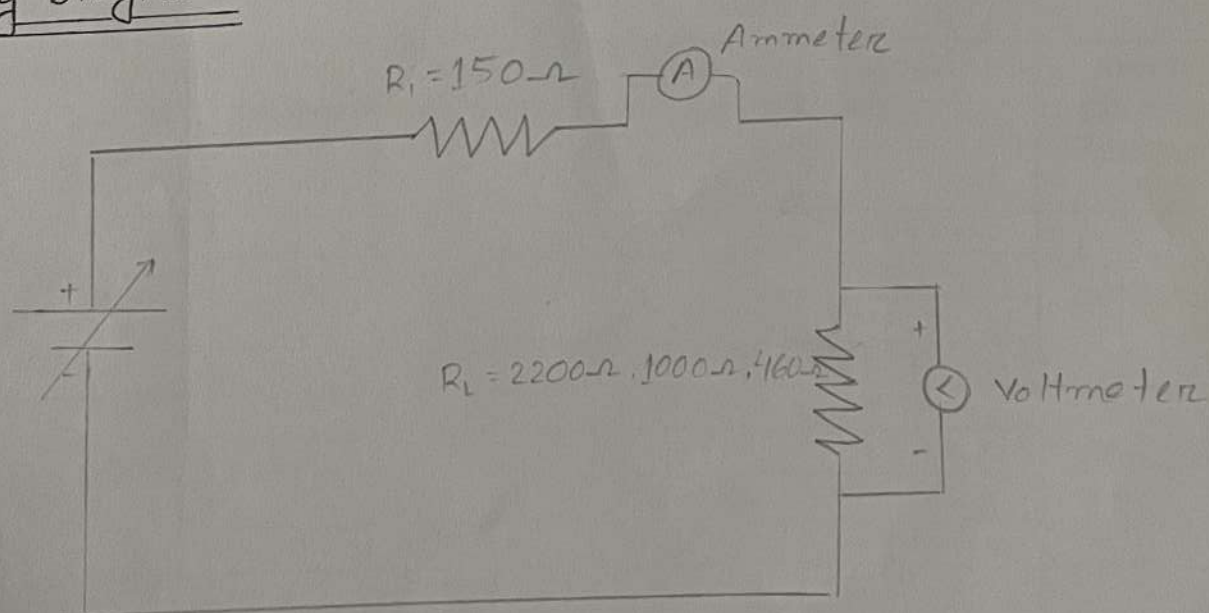
$$V = IR$$

$$\Rightarrow I = \frac{V}{R}$$

Equipment:

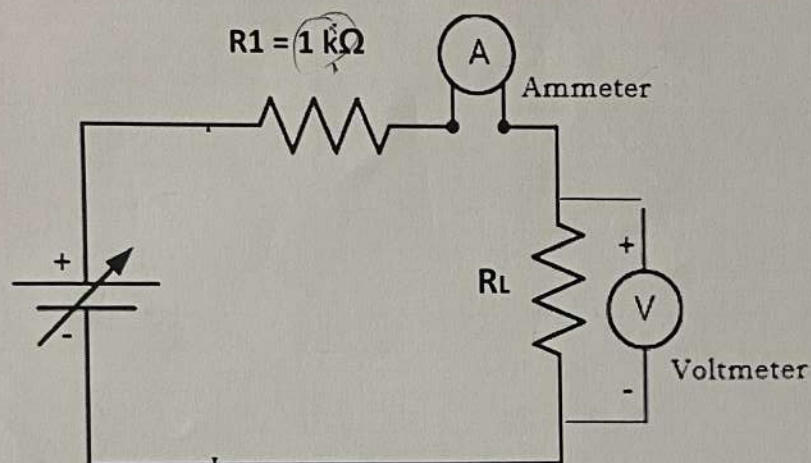
1. Variable DC power supply
2. Digital multimeter.
3. Resistance.
4. Bread Board.
5. Connecting Wires.

Working Diagram:



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Circuit Diagram:



Experimental Data:

1) Measurement of resistance with ohmmeter and check the values using color code

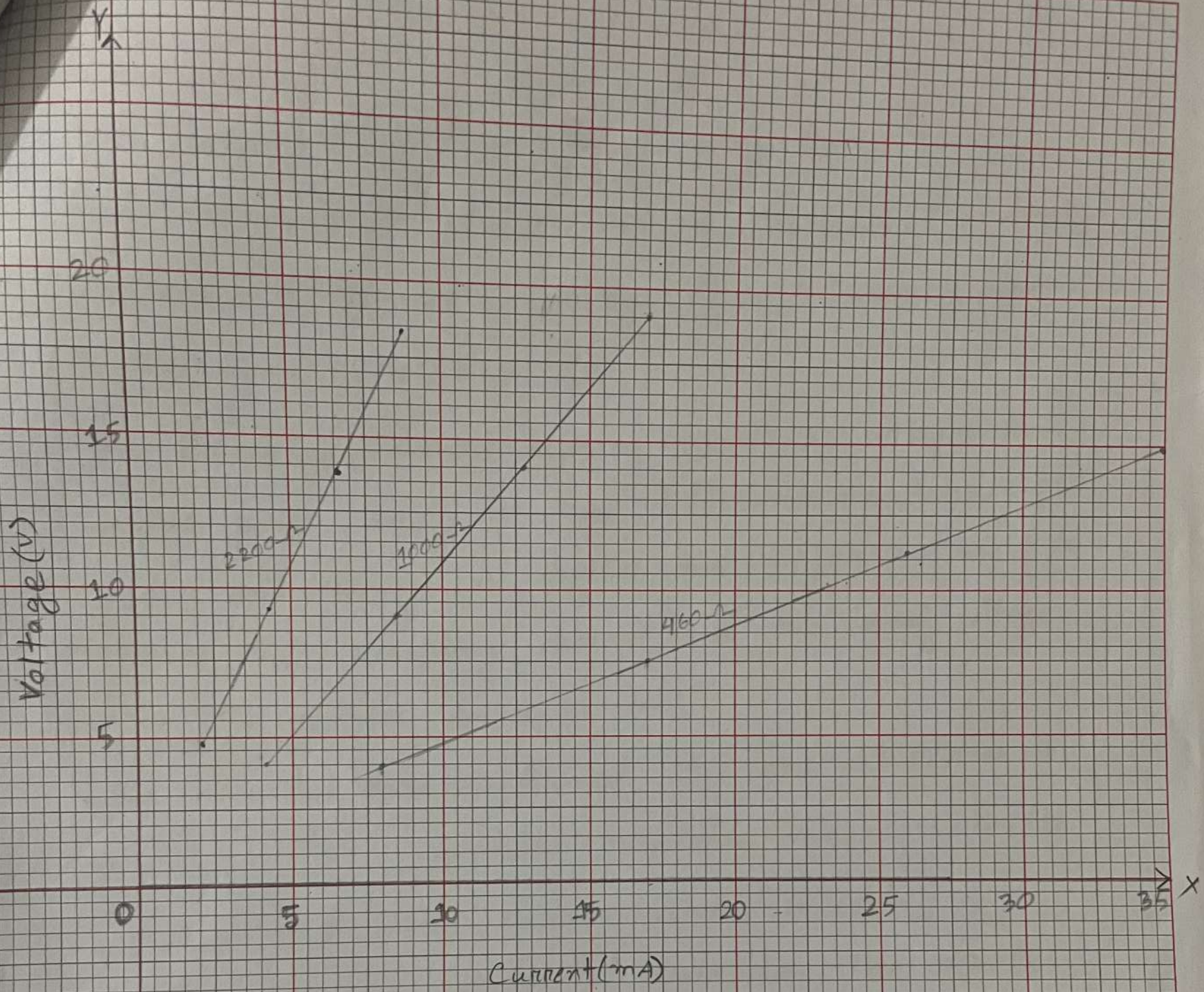
$$R_1 = 150\ \Omega$$

Resistance (kΩ) $R = R_L$	Supply Voltage (V)	Voltage (V) (across R_L)	Measured Current (I)	Calculated Resistance (R)	Average calculated Resistance (R)
2200 Ω	5	4.6 V	$2.1 \times 10^{-3}\text{ A}$	2190.47 Ω	2133.96 Ω
	10	9.2 V	$4.3 \times 10^{-3}\text{ A}$	2139.53 Ω	
	15	13.8 V	$6.6 \times 10^{-3}\text{ A}$	2090.90 Ω	
	20	18.4 V	$8.7 \times 10^{-3}\text{ A}$	2114.94 Ω	
1000 Ω	5	4.2 V	$4 \times 10^{-3}\text{ A}$	1050 Ω	953.64 Ω
	10	8.6 V	$9 \times 10^{-3}\text{ A}$	955.55 Ω	
	15	12.8 V	$14 \times 10^{-3}\text{ A}$	914.28 Ω	
	20	17 V	$19 \times 10^{-3}\text{ A}$	894.73 Ω	
460 Ω	5	3.7 V	$8 \times 10^{-3}\text{ A}$	462.5 Ω	436.89 Ω
	10	7.4 V	$17 \times 10^{-3}\text{ A}$	435.29 Ω	
	15	11.1 V	$26 \times 10^{-3}\text{ A}$	426.92 Ω	
	20	14.8 V	$35 \times 10^{-3}\text{ A}$	422.85 Ω	

Graph:

Plot a graph of voltage versus current for three different resistance values.

Signature
5/11/28



Description:

The experiment was conducted to verify Ohm's Law. This fundamental principle states that the current (I) is directly proportional to the applied voltage (V) and inversely proportional to the circuit's resistance (R).

The procedure involved applying various DC voltages from a variable power supply to three different resistors (2200Ω , 1000Ω and 460Ω). A digital multimeter was used to measure the resulting current and the actual voltage across the resistor for each trial.

The recorded data confirmed the constant relationship $R = \frac{V}{I}$. The calculated resistance values consistently matched the initial resistor values, successfully demonstrating the validity of Ohm's Law. The expected Voltage vs Current graph will show a straight line, visually confirming the direct proportionality.